

PhD-trained bioengineer with a focus on translational science, advanced in vitro models, and data-driven evaluation. Experienced in synthesizing complex biological data into clear, actionable insights supporting decision-making. Highly motivated to apply and develop my life sciences skills in biotech equity research.

PROFESSIONAL EXPERIENCE

H7 BioCapital, Business development fellow- Remote (SIE is scheduled for 7/2026) March 2026- Now

- Conducting scientific due diligence on healthcare opportunities, evaluating intellectual property, competitive landscape, and market potential to support biotech and medtech investment decisions.

Center for Discovery and Innovation (CDI)- HMH, Post-doc fellow- Genomics May 2024- Now

- Conduct deep primary research on multiple myeloma biology, integrating genomics, epigenomics, and transcriptomics across ~150 patient samples to identify prognostic biomarkers and therapeutic liabilities.
- Synthesize findings into clear written narratives for manuscripts, grant proposals, and conference presentations, translating complex science into decision-oriented insights
- Track daily literature, pipeline developments, and competitive landscape in hematologic malignancies
- Collaborate closely with clinicians, statisticians, and external academic partners, mirroring cross-functional coverage workflows

Regeneron Pharmaceuticals, Ph.D. R&D Co-Op, Human Systems Jan 2023- Aug 2023

- Supported drug discovery and preclinical evaluation using a human Liver-Chip platform to assess therapeutic efficacy, toxicity, and biomarker responses in MAFLD and MASH
- Evaluated therapeutic candidates (Lipid nanoparticle-siRNA) by assessing target gene knockdown, pathway modulation, and disease-relevant readouts
- Generated and analyzed experimental datasets used to inform go/no-go decisions
- Communicated results to internal stakeholders through concise slide decks and data summaries
- Gained exposure to industry timelines, development risk, and translational decision-making

Stevens Institute of Technology, Ph.D. Researcher- Organ-on-Chip lab Sep 2019- May 2025

- Led multiple research programs investigating mechanisms of drug resistance directly relevant to clinical outcomes and therapeutic durability in hematologic malignancies
- Designed physiologically relevant disease models to evaluate therapeutic response under clinically realistic conditions in multiple myeloma
- Integrated transcriptomic, proteomic, and functional data to support target prioritization
- Regularly presented findings to technical and non-technical audiences, strengthening investment-style communication skills

EDUCATION

Stevens Institute of Technology, Hoboken, NJ, Ph.D., Biomedical Engineering May 2025

Thesis: "Optimizing multiple myeloma treatment approaches by elucidating tumor-stroma interactions in the bone marrow microenvironment." Identified key signaling pathways significantly enhancing drug resistance in MM within the BME and optimized therapeutic approaches (*Behboodi, et al., in preparation*).

Sharif University of Technology, Tehran, Iran, M.Sc., Materials Science and Engineering May 2019

Thesis: "Design, Fabrication, and Improvement of Mechanical Properties and Biocompatibility of Hydroxyapatite-Chitosan Scaffold for Bone Tissue Engineering Applications."

Sharif University of Technology, Tehran, Iran, B.Sc., Materials Science and Engineering May 2017

Thesis: "Development and Mechanical Properties of Bimodal Microstructures in Low Carbon Steel."

KEY SKILLS

Experienced in conducting primary scientific research across oncology and immunology, including evaluation of disease biology, mechanism-of-action, therapeutic modalities, targets, and biomarkers. Skilled in analyzing and integrating complex biological datasets (NGS, RNA-seq, epigenomics) using R and Python, and in mining public resources such as TCGA and GEO to identify drivers of therapeutic response or resistance. Strong translational insight across drug development, including assessment of preclinical efficacy signals, patient-stratification strategies, and targetable pathway dependencies.

HONORS AND AWARDS

- **Awarded \$50k** for a one-year research project titled “Understanding the Interactions Between Multiple Myeloma Cells and Osteoblasts to Optimize Treatment Approaches” by New Jersey Health Foundation (Grant # PC 238-25).
- **Runner-Up**, 1-minute Thesis Pitch Award – [Johnson & Johnson Engineering Showcase 2025](#)
- Recipient of the **Provost Fellowship**, the most prestigious PhD scholarship of Stevens Institute of Technology.

PUBLICATIONS

- Alshehri S, Pavlovič T, Farsinejad S, **Behboodi P**, et al., Extracellular Matrix Modulates Outgrowth Dynamics in Ovarian Cancer. [Advanced Biology](#) 2022.
- Arab M, **Behboodi P**, et al., Enhanced Mechanical Properties and Biocompatibility of Hydroxyapatite Scaffolds by Magnesium and Titanium Oxides for Bone Tissue Applications. [Heliyon](#) 2024.
- Iwanicki M, Pavlovic T, Behboodi P, Engineering Approaches in Ovarian Cancer Cell Culture. Engineering and Physical Approaches to Cancer. Current Cancer Research. [Springer, Cham](#) 2023.
- Bokaii Hosseini Z, Rajabi F, Morovatshoar R, Ashrafpour M, **Behboodi P**, et al., Downregulation of LPAR1 Promotes Invasive Behavior in Papillary Thyroid Carcinoma Cells. [Cancer Informatics](#) 2024.
- Aziz M, **Behboodi P**, Lee W, Assessing Natural Killer Cell Cytotoxicity Against Multiple Myeloma Cells with IL-6 and sIL-6R. [Journal of Student Research](#) 2024.

CONFERENCES and WORKSHOPS

- **Behboodi P**, et al., *MACROD2* and *CDKN2A* in multiple myeloma: insights to germline susceptibility and cytogenetic risk from long-read Nanopore sequencing, [ASCO Annual meeting](#), Chicago, May 2026.
- **Behboodi P**, et al., Effects of PFAS on DNA methylation at human cancer susceptibility loci, [ASHG Annual meeting](#), Boston, October 2025.
- **Behboodi P**, et al., Optimizing multiple myeloma treatment approaches by elucidating tumor-stroma interactions in the bone marrow microenvironment, [AACR Annual Meeting](#), Chicago, IL, April 2025.
- **Behboodi P**, et al., Optimizing Multiple Myeloma Treatment, [Johnson & Johnson Engineering showcase](#), (Poster presentation), New Brunswick, NJ, April 2025.
- Chim S M, Howell K, ..., **Behboodi P**, et al., Development of Human Systems for genetic disease modeling, drug development and new biology, Human Systems, VelociGene. [Regeneron Pharmaceuticals Inc.](#), Tarrytown, NY 2023.
- Castano A, Rigo M, ..., **Behboodi P**, et al., Allele-specific methylation mapping and CRISPR mutagenesis identify functional chromatin-regulating SNPs (rSNPs) in disease risk loci, [Genetics and Epigenomics research conference](#), Center for Discovery and Innovation (CDI) associated with [Hackensack Meridian Health \(HMH\)](#), NJ 2024.
- Alshehri S, Pavlovič T, ..., **Behboodi P**, et al., Extracellular Matrix Modulates Outgrowth Dynamics in Ovarian Cancer, [48th Annual Northeast Bioengineering Conference](#), Columbia University, NY 2022.
- Science Alliance Leadership Training (SALT) Workshop, [New York Academy of Sciences](#), NY 2022. Selected to participate in a comprehensive workshop focused on enhancing leadership skills.